



Module 6: Introducing Topic Modeling

AWS Academy Natural
Language Processing

Module overview

- Sections
 - Section 1: Introduction to topic modeling
 - Section 2: Identifying the approach
 - Section 3: Implementing topic modeling
- Labs
 - Guided Lab: Implementing Topic Modeling with Amazon Comprehend
 - Guided Lab: Implementing Topic Modeling Using Neural Topic Model (NTM)
 - Challenge Lab: Implementing Topic Modeling

Module objectives

- At the end of this module, you should be able to:
- Understand the use cases for topic modeling
- Describe the steps for topic modeling
- Use Amazon Comprehend to implement a topic modeling solution
- Evaluate machine learning (ML) algorithms that are used in natural language processing (NLP) for topic modeling
- Create a solution to a topic modeling business problem

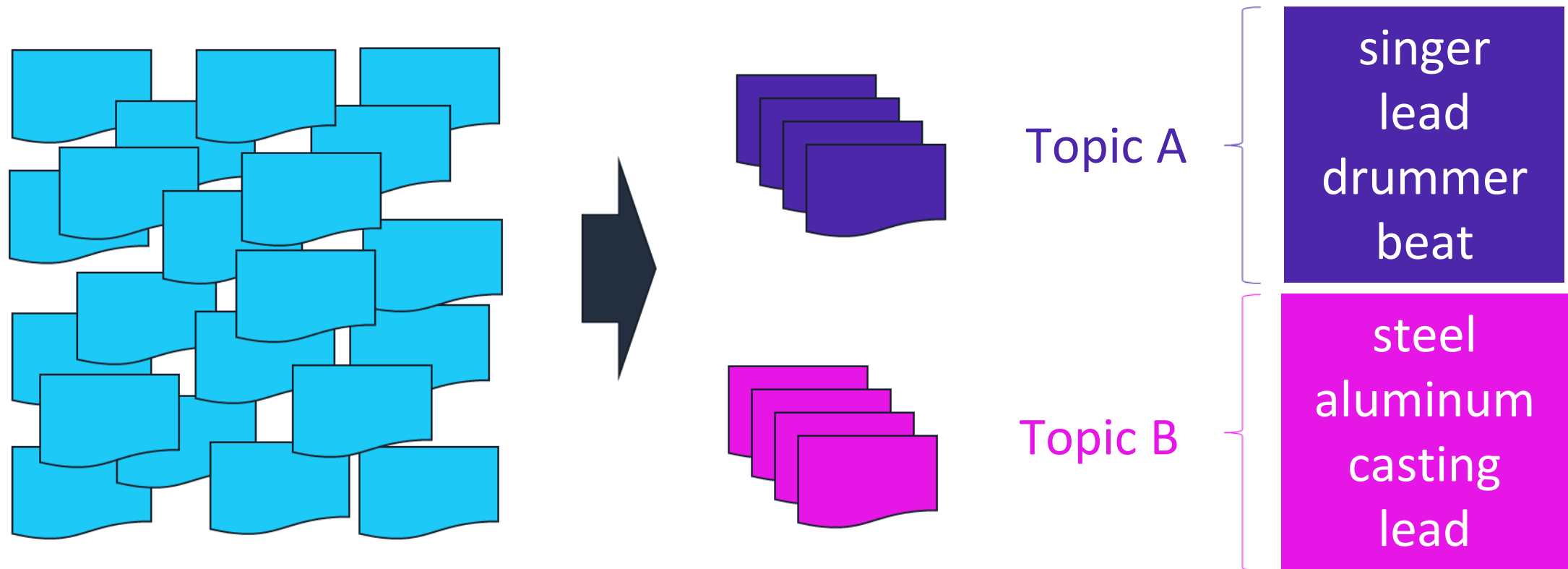


Section 1: Introduction to topic modeling

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What is topic modeling?

Finding topics in a large collection of documents



Topic modeling use cases

- Document classification
- Automatic content tagging
- Document summarization
- Information retrieval by using topics
- Content recommendation based on topics

Section 1: Summary

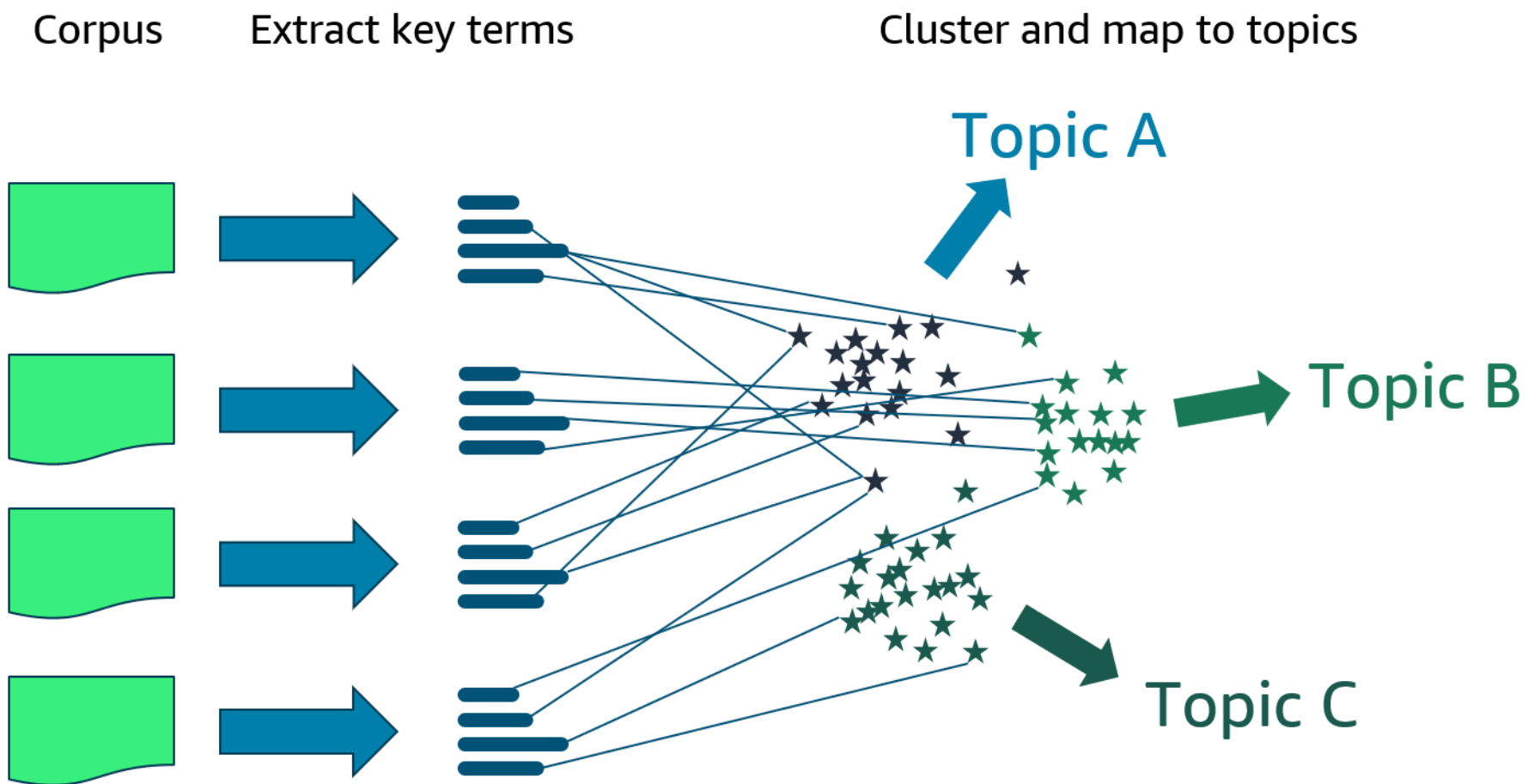
- What is topic modeling?
- Topic modeling use cases:
 - Document classification
 - Automatic content tagging
 - Document summarization
 - Information retrieval by using topics
 - Content recommendation based on topics



Section 2: Identifying the approach

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Topic modeling approaches



Processing steps

- Clean the data
- Remove stopwords
- Lemmatize the data
- Vectorize or tokenize the documents

Section 2: Summary

- Topic modeling approaches
- Processing steps
 - Clean the data
 - Remove stopwords
 - Lemmatize the data
 - Vectorize and tokenize

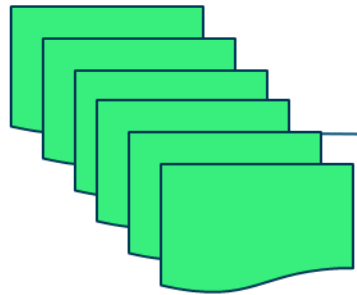


Section 3: Implementing topic modeling

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Amazon Comprehend for topic modeling

>1,000 documents



Documents in Amazon
Simple Storage Service
(Amazon S3)



Amazon Comprehend

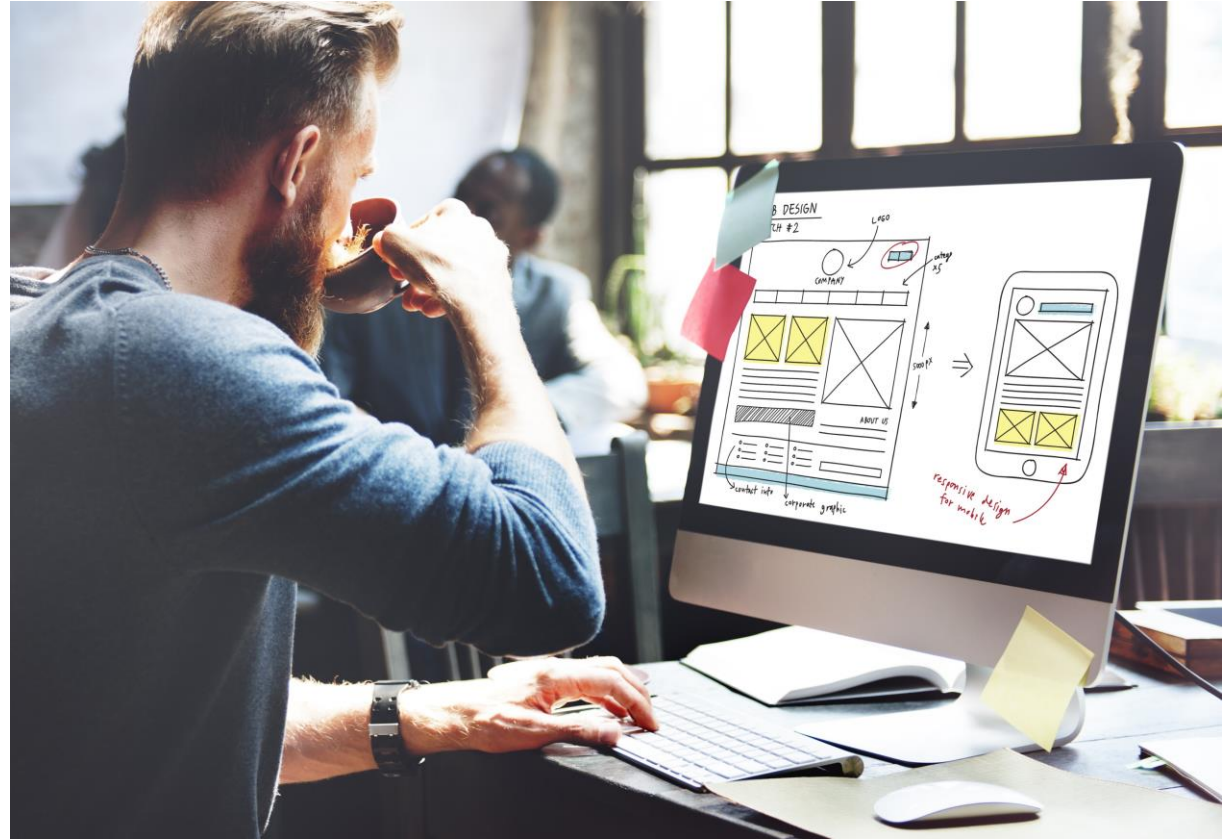
topic-terms.csv

Topic	Term	Weight
000	team	0.118533
000	game	0.106072
001	spoon	0.031625
001	pepper	0.022205

doc-topics.csv

Document	Topic	Proportion
doc1	000	0.9993
doc2	001	0.9985
doc3	002	0.9953

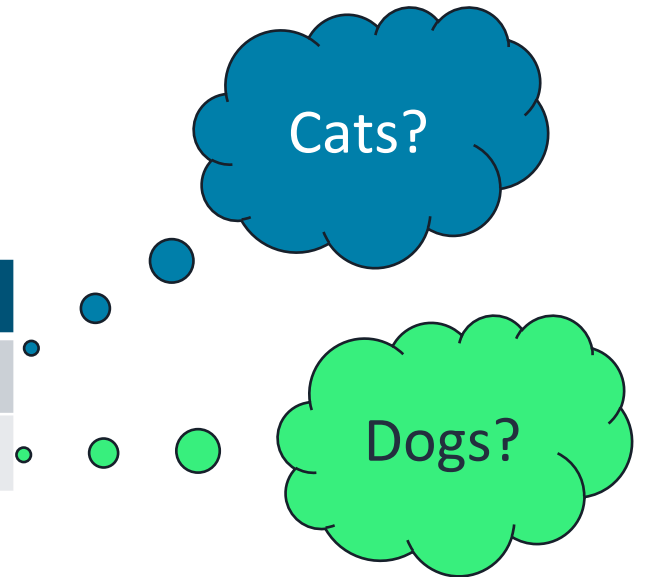
Module 6 Guided Lab 1: Implementing Topic Modeling with Amazon Comprehend



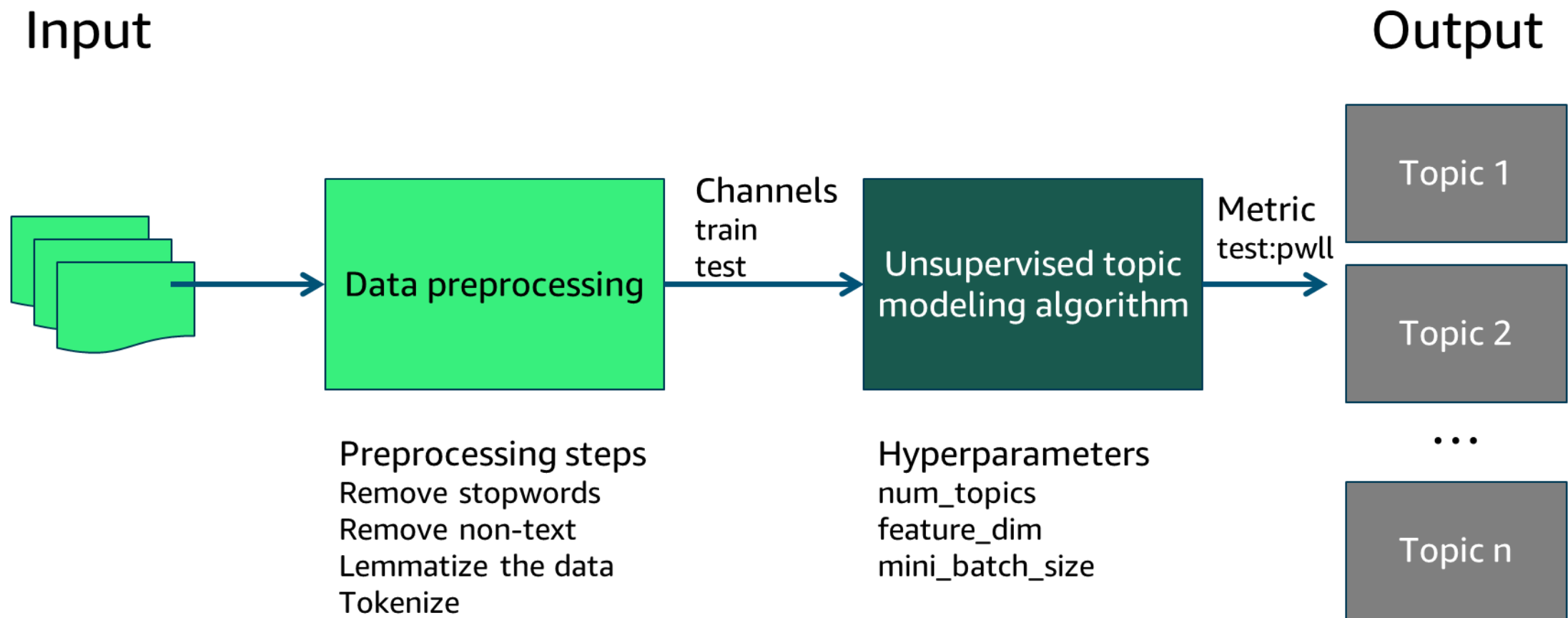
Latent Dirichlet Allocation (LDA)

- Unsupervised learning algorithm
- Describes a set of documents as a mixture of distinct topics
- Topics are NOT specified up front
- Might not align with how a human would categorize
- Word groups = topics

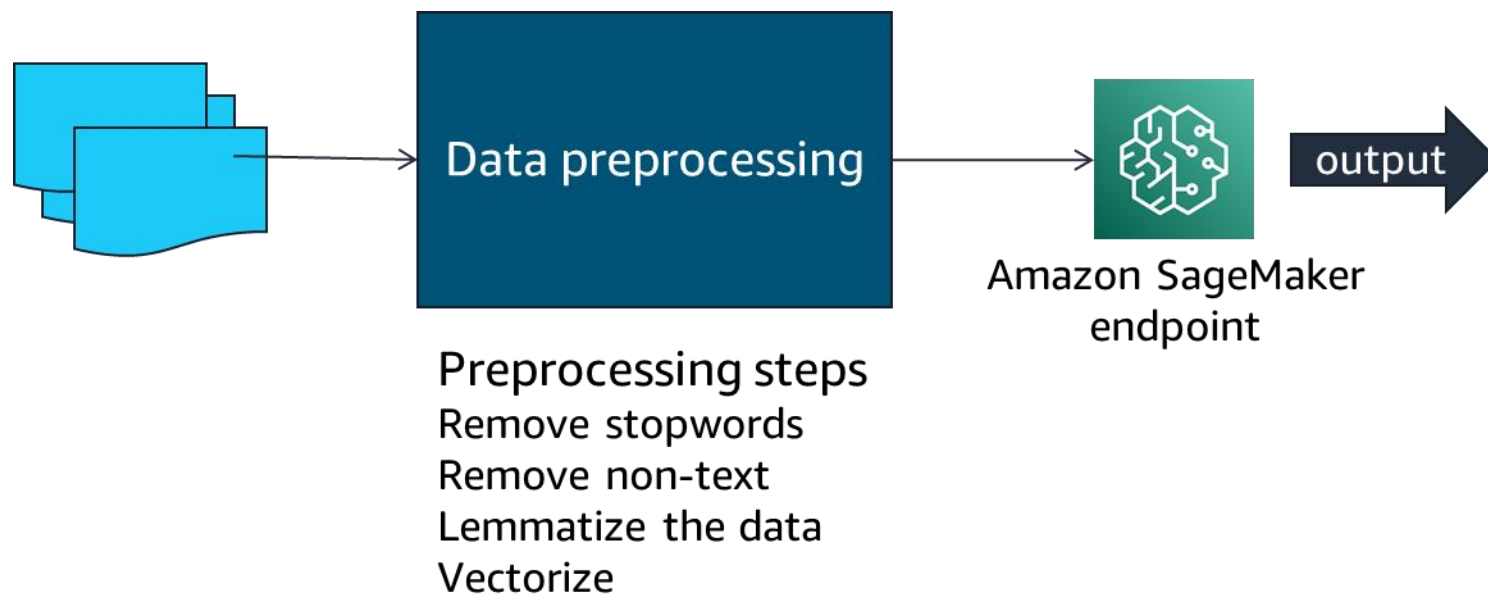
Topic	eat	sleep	play	meow	bark
Topic 1	0.1	0.3	0.2	0.4	0.0
Topic 2	0.2	0.1	0.4	0.0	0.3



Training an LDA model



Using a trained LDA model

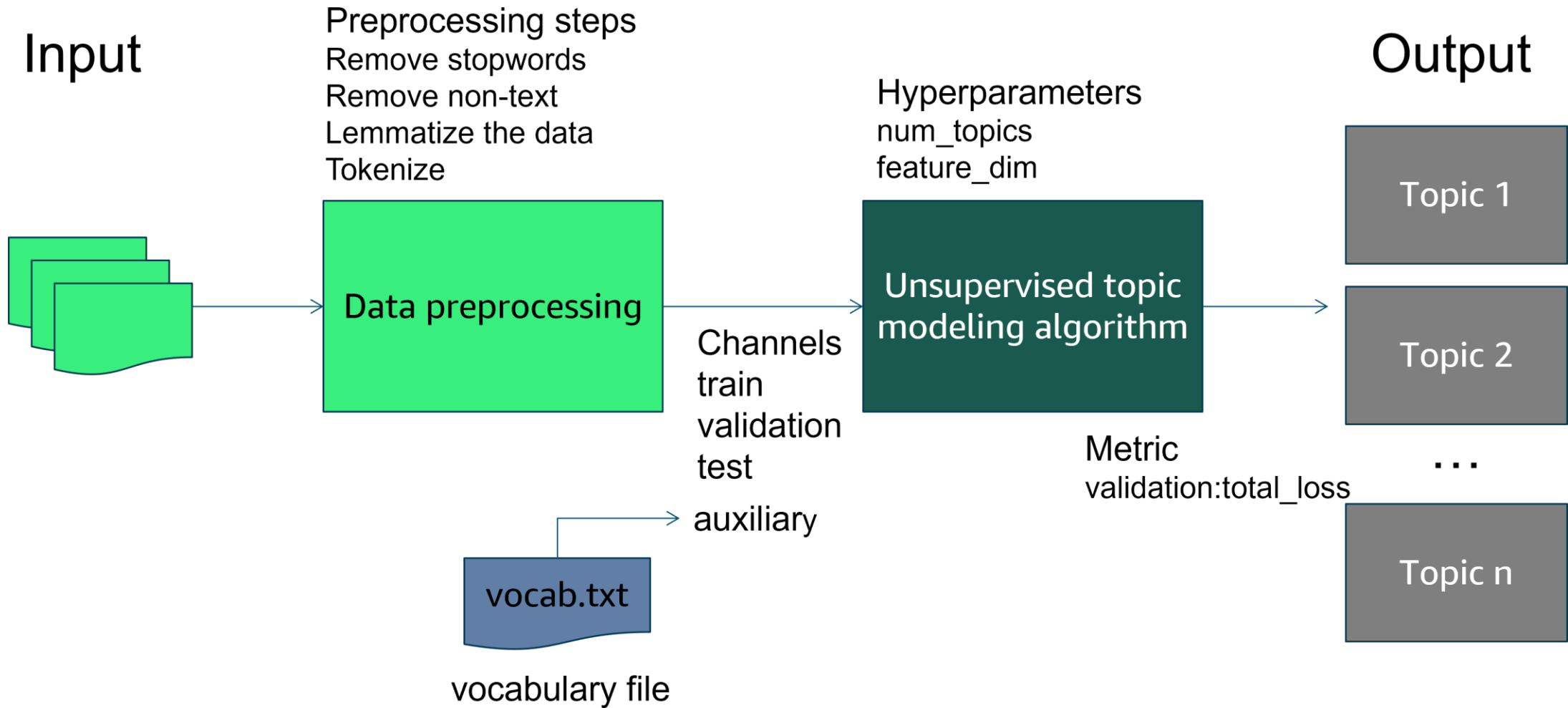


```
{
  "predictions": [
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      "topic_mixture": [
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        0.05404851585626602,
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    },
    {
      "topic_mixture": [
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        0.0,
        0.2668071687221527,
        0.0
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    }
  ]
}
```

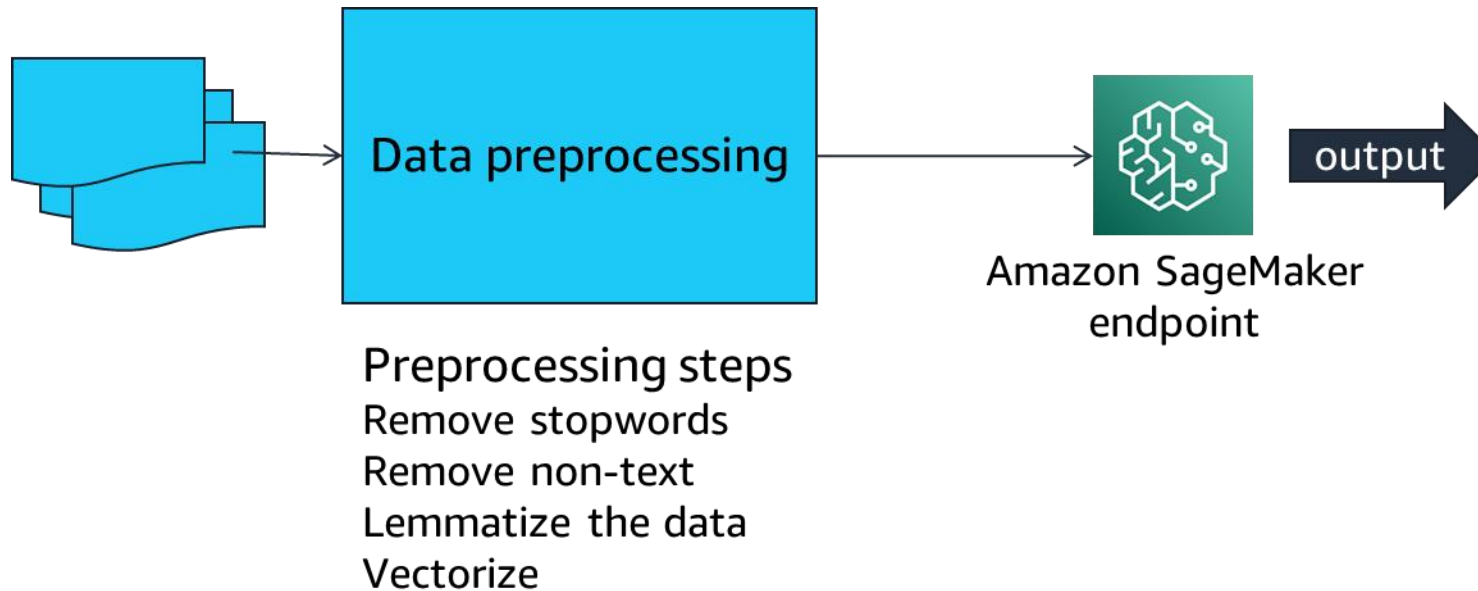
Neural Topic Model (NTM) algorithm

- Unsupervised learning algorithm
- Organizes a corpus of documents into *topics* that contain word groupings based on their statistical distribution
- Topics are NOT specified up front
- Might not align with how a human would categorize
- Can produce different results than LDA

Training an NTM model



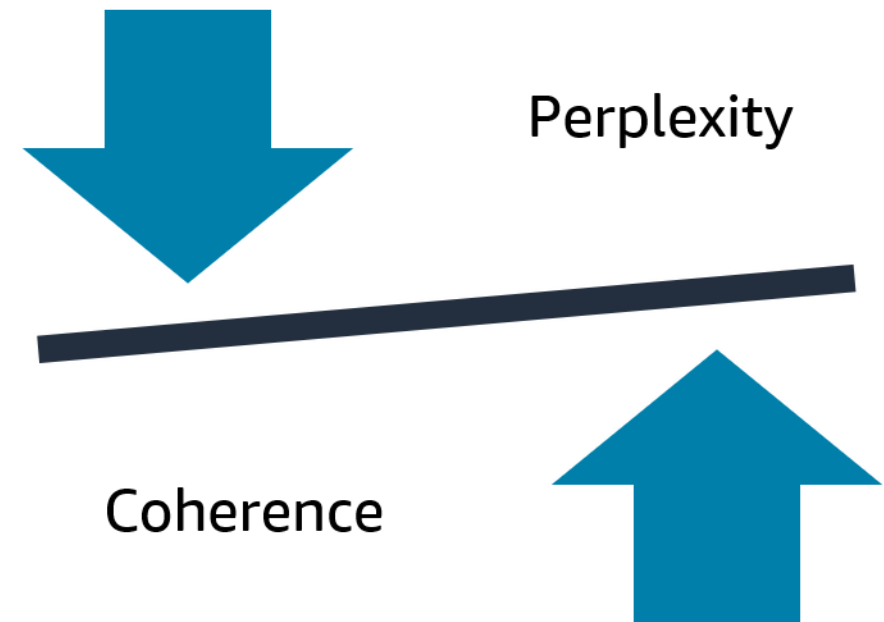
Using NTM



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      ]
    },
    {
      "topic_weights": [
        0.05667472630739212,
        0.6765181422233582,
        0.0,
        0.2668071687221527,
        0.0
      ]
    }
  ]
}
```

Perplexity and coherence

- **Perplexity:** Metric for the likelihood that a topic is included in a document
- **Coherence:** Metric for the similarity of words in a topic
- Metrics are useful to evaluate topic models
 - Lower perplexity scores = model correctly identifies the subject
 - Higher coherence = topics in the document are similar



Choosing between Amazon Comprehend, LDA, and NTM

Amazon Comprehend

- Fully managed service
- Easier if you are new to machine learning
- Tight deadlines

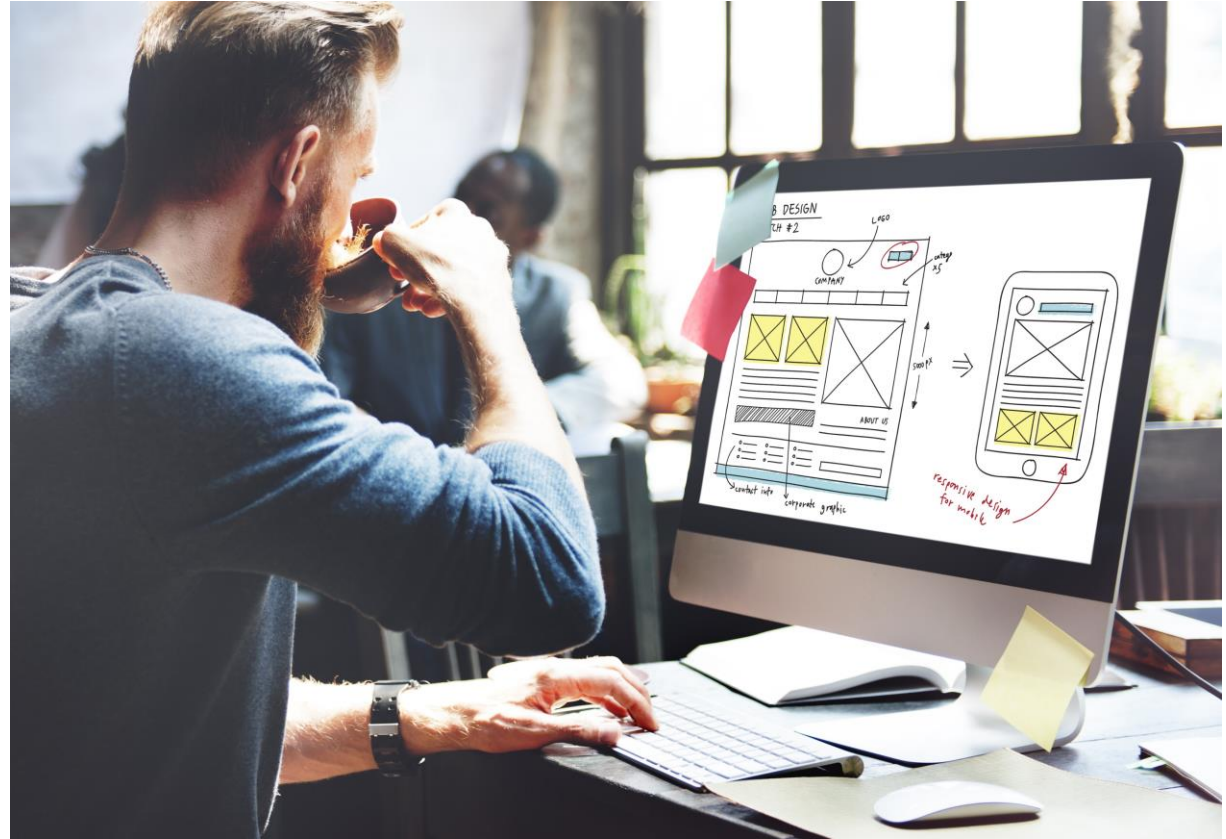


Amazon SageMaker built-in algorithms

- Require training
- Customized vocabulary
- Regulatory hurdles

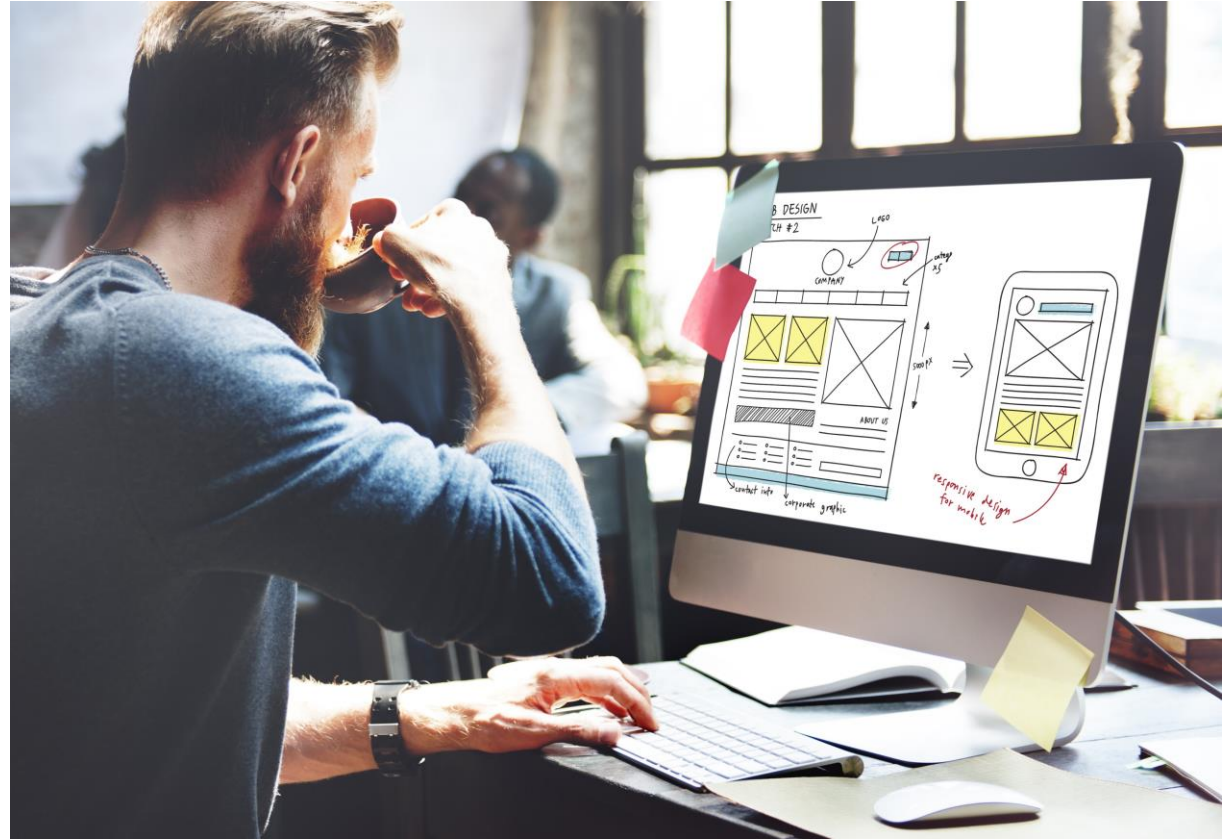


Module 6 Guided Lab 2: Implementing Topic Modeling Using Neural Topic Model (NTM)



Module Challenge

Lab 3: Implementing Topic Modeling



Section 3: Summary

- Using Amazon Comprehend
- Using Amazon SageMaker algorithms
 - LDA
 - NTM
- Choosing between Amazon Comprehend, LDA, and NTM



Thank you

Corrections, feedback, or other questions?

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